

LAWTON FORT SILL NORTH, OK, USA

WMO: 723550

Lat: 34.650N

Lon: 98.400W

Elev: 362

StdP: 97.05

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13945

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -6.6	(d) -16.3	(e) 0.9	(f) -6.4	(g) -13.6	(h) 1.2	(i) -3.8	(j) 13.2	(k) 2.7	(l) 12.0	(m) 3.0	(n) 4.8	(o) 10	(p) 0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.7	(c) 39.8	(d) 22.7	(e) 38.0	(f) 23.1	(g) 36.5	(h) 23.2	(i) 26.5	(j) 33.6	(k) 25.7	(l) 32.7	(m) 25.0	(n) 32.0	(o) 4.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.6	30.4	23.9	19.6	29.2	23.1	18.6	28.2	85.1	33.5	81.1	32.5	78.1	32.0	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.7	(b) 10.4	(c) 9.1	(e) -13.0	(f) 41.2	(g) 2.5	(h) 2.0	(i) -14.8	(j) 42.7	(k) -16.3	(l) 43.9	(m) -17.7	(n) 45.0	(o) -19.5	(p) 46.5		
(4) 11.7	(5) 10.4	(6) 9.1	(7) -13.8	(8) 26.9	(9) 2.2	(10) 1.2	(11) -15.3	(12) 27.8	(13) -16.6	(14) 28.5	(15) -17.9	(16) 29.2	(17) -19.5	(18) 30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	4.9	7.4	12.0	16.5	21.5	26.9	29.1	28.5	24.3	17.7	11.1	5.7
	DBStd	9.71	5.56	6.18	5.75	4.49	4.17	3.05	2.84	2.96	4.12	4.76	4.83	5.16
	HDD10.0	531	172	109	44	6	0	0	0	0	0	5	44	149
	HDD18.3	1719	416	307	205	85	18	0	0	0	6	69	221	391
	CDD10.0	3156	14	37	108	201	356	507	592	574	431	243	76	17
	CDD18.3	1302	0	2	10	29	116	257	334	316	187	49	3	0
	CDH23.3	15783	5	30	124	337	1159	3086	4513	4014	1982	490	39	3
	CDH26.7	8333	0	6	28	95	461	1622	2676	2309	977	154	4	0
Wind	WSAvg	4.5	4.5	4.8	5.4	5.3	4.7	4.6	3.9	3.8	3.9	4.3	4.4	4.3
Precipitation	PrecAvg	738	28	33	56	64	112	97	52	63	84	72	46	37
	PrecMax	1060	82	117	173	158	415	261	214	240	190	281	155	148
	PrecMin	417	0	0	0	2	4	16	0	0	9	4	0	0
	PrecStd	189	22	26	43	43	82	63	46	56	52	68	40	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	26.8	29.8	32.3	36.1	40.0	42.1	41.9	38.8	33.1	27.2	22.9
		MCWB	13.2	14.7	16.8	17.9	21.0	22.6	22.7	22.2	21.8	21.1	17.6	14.0
	2%	DB	19.9	23.5	26.6	29.3	33.5	37.5	40.0	39.5	36.2	30.3	24.1	19.6
		MCWB	11.3	13.8	16.9	18.0	22.1	22.9	22.6	23.4	22.5	20.1	16.4	13.2
	5%	DB	17.1	20.2	24.2	27.2	31.3	35.6	38.1	37.7	34.2	28.1	21.7	16.8
		MCWB	10.2	13.0	15.7	17.6	21.6	23.2	23.1	23.5	22.7	19.3	14.8	11.5
	10%	DB	14.4	17.4	21.7	25.0	29.3	33.9	36.6	35.9	32.2	26.1	19.3	14.1
		MCWB	8.5	11.1	14.4	17.0	21.2	23.8	23.4	23.3	22.2	18.1	13.7	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	18.5	20.5	22.4	25.7	27.5	27.5	27.5	26.4	24.4	19.9	17.8
		MCDB	20.5	21.4	25.4	27.3	30.7	33.6	34.6	36.5	31.8	28.6	23.3	19.8
	2%	WB	13.2	16.3	19.0	20.9	24.3	26.2	26.4	26.1	25.2	22.5	18.0	14.9
		MCDB	17.1	20.1	23.5	25.4	29.2	32.6	34.0	33.7	31.5	27.2	21.7	17.3
	5%	WB	10.9	14.3	17.4	19.9	23.2	25.2	25.5	25.2	24.2	21.1	16.5	12.3
		MCDB	15.9	18.1	21.9	24.3	28.7	31.8	33.1	32.6	30.1	25.1	20.0	15.5
	10%	WB	8.8	12.0	15.7	18.8	22.1	24.4	24.8	24.5	23.3	19.8	14.9	9.8
		MCDB	13.9	17.3	20.4	23.0	27.5	31.2	32.2	31.7	29.0	24.4	18.5	13.8
Mean Daily Temperature Range	5% DB	MDBR	13.0	12.8	13.1	13.0	11.9	12.0	12.7	12.6	12.2	13.0	12.8	11.5
		MCDBR	17.8	17.8	16.4	16.3	14.4	14.3	15.1	14.9	14.8	15.3	16.0	16.2
		MCWBR	10.5	10.0	8.7	8.1	5.8	4.5	3.8	3.9	4.7	7.0	9.4	10.0
	5% WB	MCDBR	15.8	15.1	13.7	12.7	12.1	12.0	12.7	12.5	11.9	11.9	13.4	13.6
		MCWBR	10.1	10.2	8.4	7.9	6.3	5.1	4.7	4.4	4.9	7.1	9.4	10.0
Clear-Sky Solar Irradiance	taub		0.289	0.298	0.335	0.369	0.405	0.425	0.421	0.417	0.384	0.347	0.311	0.292
	taud		2.520	2.511	2.423	2.353	2.323	2.306	2.352	2.307	2.373	2.466	2.500	2.511
	Ebn at Noon		927	955	939	916	882	860	862	860	874	885	891	901
	Edh at Noon		81	92	109	123	129	131	124	128	114	95	83	78
All-Sky Solar Radiation	RadAvg		2.95	3.54	4.70	5.75	6.19	6.87	6.91	6.32	5.25	4.03	3.10	2.50
	RadStd		0.30	0.50	0.30	0.46	0.66	0.60	0.41	0.37	0.42	0.56	0.39	0.28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page